

PROFESSIONAL SUMMARY

- Backend engineer with **4+ years** building scalable distributed systems, with deep expertise in **database performance tuning, zero-downtime schema migrations, and Ruby on Rails** in large production environments.
- Delivered production-critical database wins: **208× query speedup** on a 524M-row table, zero-downtime primary-key migrations on **3 tables of 1B+ rows** (~1TB), and **4-region** rollouts — all without downtime.
- Experienced in **remote-first, async collaboration**; open-source contributor (GSoC 2022, GNOME Foundation) with strong alignment to transparent, distributed engineering culture.

EXPERIENCE

- **BrowserStack** Mumbai
Backend SDE - *Ruby on Rails, Java/Spring Boot, Node.js, ClickHouse, Kafka, AWS* Apr 2025 – Present
 - Drove a **208× query speedup** (9.6 s → 46 ms) on a **524M-row** table via a **zero-downtime** schema migration across **4 regions**.
 - Averted an outage via **zero-downtime** primary-key migrations on **3 tables of 1B+ rows** (~1TB) growing **2.9M rows/day**.
 - Cut **database CPU from 96%** to stable by offloading heavy aggregate queries to a columnar store (**ClickHouse**).
 - Cut dashboard load time from **3–4 min** → **3 s** via indexed queries and query plan analysis, unblocking a stalled enterprise deal.
 - Launched a new **AWS region (Australia)**, now serving **360K requests/hour** in production; coordinated multi-region data migration and cutover.
 - Built the **access-control system** from the ground up — **RBAC** with **multi-team** support across **~14 enforcement points**.
 - Remediated **SQL injection** across **50+ APIs** and rolled out **CI security scanning** covering **~98% of traffic**.
- **Zus (formerly 0chain)** Cupertino, CA (Remote)
Software Engineer - *Blockchain Storage & Mobile - Golang, Kotlin, GitHub Actions, Gin, Grafana* Feb 2022 – Mar 2025
 - Shipped 2 production Android apps, **Bolt** (crypto wallet) and **Vult** (storage), driving **20%** adoption growth and **\$2M+** in transactions.
 - Built a **UAT automation suite** for blockchain storage, cutting bugs by **90%** and release cycles from **2 weeks to 3 days**.
 - Optimized **S3-compatible storage** handling **500M+ files** and **40TB+** across **50+ nodes** at sub-**100ms** latency, **99.9%** uptime.
 - Automated dynamic **Sharder/Miner** management and view-change protocols, boosting developer productivity by **90%**.
 - Enhanced **multi-wallet concurrency** via unified **GoSDK**, achieving **3× throughput** improvement under load testing.
- **Oppia Foundation** (incl. **GSoC**) [**Merged PRs**] Mountain View, CA (Remote)
OSS Contributor & GSoC - *Kotlin, GitHub Actions, JUnit, Gradle, Bazel, Android SDK* Jun 2021 – Sep 2022
 - Contributed to **Gradle to Bazel** migration, achieving **10x** improvement in build speeds.
 - Built **CI/CD pipelines** (GSoC) automating build triggers and deployment to **Google Drive**, **GCP Bucket**, and **Play Store** tracks.
 - Developed modular **MVVM/MVP** architectures with **tablet support** for multiple screen sizes.
 - Authored **70+ Unit Tests** and **20+ E2E tests** using Espresso, Robolectric, and UIAutomator.
- **GNOME Foundation** Remote
Open Source Contributor - *Tech Stack: C, XML, GTK+* Sep 2021 – Feb 2022
 - Enhanced **Nautilus file manager** with new path-bar features enabling faster navigation without leaving the keyboard.
 - Implemented **drag-and-drop support** and keyboard shortcuts for seamless file organization.

PROJECTS

- **Zus HelloWorld (Android)**: Reference **Kotlin** app demonstrating **Zus/0chain GoSDK** integration — **Bolt** crypto wallet (ZCN/Ethereum staking) and **Vult** decentralized storage with multi-file upload, download, and thumbnail generation.

ACHIEVEMENTS

- Selected for **Google Summer of Code 2022** (Oppia Foundation) — merged **open-source** CI/CD tooling used in production.
- Won **top prize (16K INR)** at the **CAMS Hackathon** for developing its Android app.
- Rated **Specialist** on **Codeforces** (max-rating **1509**) and **5-star** on **CodeChef**.

TECHNICAL SKILLS

- **Programming & Scripting:** Ruby, Java, Go, Kotlin, Python, JavaScript, SQL, Bash, C/C++
- **Backend Frameworks:** Ruby on Rails, Spring Boot, Gin, Express.js, REST APIs, GraphQL, Microservices
- **DevOps & Cloud:** AWS (RDS), Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD, Helm, Prometheus/Grafana/Loki
- **Databases & Testing:** PostgreSQL, MySQL, MongoDB, Redis, ClickHouse, Query Optimization, Schema Migrations, JUnit, Mockito
- **Tools & Libraries:** Git, Apache Kafka, Change Data Capture (CDC), Zero-downtime Migrations, Static Analysis (SAST)

EDUCATION

- **Indian Institute of Information Technology, Lucknow** Lucknow
Bachelor of Information Technology; CGPA: 8.32 Apr 2020 – Apr 2024